Generative Permutations

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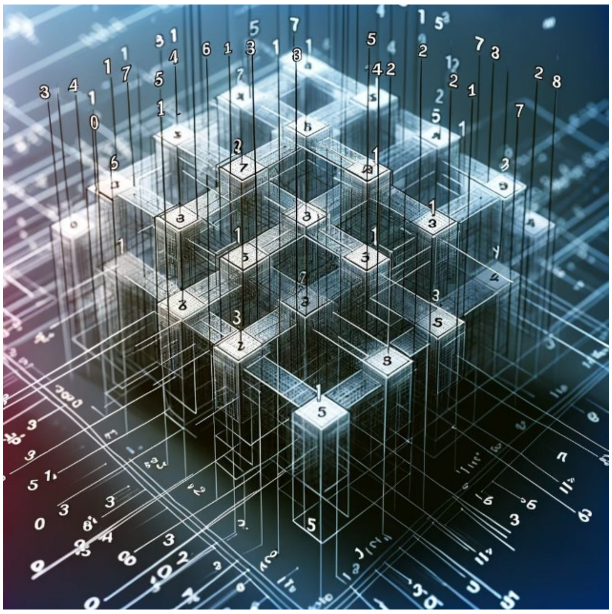
Problem Statement:



A permutation A of length n is a list of n numbers a_0, a_1,..., a_{n-1}. The numbers in the list are the integers from 0 to n-1, and there are no repetitions. The permutation is defined as generative if there exists another list B formed by b_0, b_1,..., b_{n-1} defined by $b_i=(a_i+i) \bmod n$ for every i such that B is a permutation of the numbers in A.

Given n integer numbers from 0 to n-1, your task is to determine if these numbers form a permutation and, if so, whether this permutation is generative.

Input / Output

Input Format:

The input will consist of a sequence of integers separated by spaces.

Output Format:

The output will be: ERROR, if the input is not composed of integer numbers or the n numbers in the input do not range from 0 to n-1.

The output will be: NOT A PERMUTATION, if the n numbers do not form a permutation.

The output will be: IS A GENERATIVE PERMUTATION, if the permutation is generative, or NOT A GENERATIVE PERMUTATION, otherwise.

Input / Output

Example 1

Input:

```
1 2 0
```

Output:

```
ES UNA PERMUTACIÓN GENERATIVA
```

Example 2

Input:

```
0 1 2 4 6 3 8 5 7
```

Output:

```
NO ES UNA PERMUTACIÓN GENERATIVA
```

Example 3

Input:

```
1 2 3
```

Output:

```
ERROR
```